



PDF Download
3732365.3732409.pdf
02 March 2026
Total Citations: 3
Total Downloads: 867

Latest updates: <https://dl.acm.org/doi/10.1145/3732365.3732409>

RESEARCH-ARTICLE

Anomalous Network Access Detection with Large Language Model

LEI ZHANG, State Grid Corporation of China, Beijing, China

JIAXUAN WU, Tongji University, Shanghai, China

NAIWANG GUO, State Grid Corporation of China, Beijing, China

BINBIN WANG, State Grid Corporation of China, Beijing, China

YI WU, State Grid Corporation of China, Beijing, China

MENGYUAN ZHANG, State Grid Corporation of China, Beijing, China

[View all](#)

Open Access Support provided by:

State Grid Corporation of China

Tongji University

Published: 26 June 2025

Citation in BibTeX format

CNSSE 2025: 2025 5th International
Conference on Computer Network
Security and Software Engineering
February 21 - 23, 2025
Qingdao, China

Anomalous Network Access Detection with Large Language Model

Lei Zhang
The State Grid Shanghai Municipal
Electric Power Company
Shanghai, China
lei.zhang8@mail.mcgill.ca

Jiaxuan Wu
Tongji University
Shanghai, China
2432227@tongji.edu.cn

Naiwang Guo
The State Grid Shanghai Municipal
Electric Power Company
Shanghai, China
guonw@foxmail.com

Binbin Wang
The State Grid Shanghai Municipal
Electric Power Company
Shanghai, China
wangbinbin1129@163.com

Yi Wu
The State Grid Shanghai Municipal
Electric Power Company
Shanghai, China
wu.yi.christian@gmail.com

Mengyuan Zhang
The State Grid Shanghai Municipal
Electric Power Company
Shanghai, China
z_mengyuan@163.com

Yinxia Luan*
Tongji University
Shanghai, China
22086@tongji.edu.cn

Abstract

Accurately detecting and handling abnormal access behaviors is crucial for the stable operation of network systems, while traditional anomaly detection methods are usually difficult to deeply understand and identify existing complex network security scenarios and potential threats, which limits their application in power network systems. To solve this problem, this paper proposes a network anomaly detection method based on a large language model (LLM). This method constructs a high-quality network security dataset by aligning multimodal network system data with natural language methods, and uses continuous pre-training and supervised fine-tuning methods to train a vertical domain large language model (NS-LLM) for network system anomaly detection, thereby helping the vertical domain model to effectively understand complex network access behavior patterns. In addition, this method also uses the model's network security knowledge and cross-domain reasoning capabilities to construct a mechanism for identifying and handling abnormal patterns, further enhancing the system's ability to detect and respond to complex or potentially abnormal access behaviors. Experimental results show that the method designed in this paper is significantly superior to traditional methods in terms of accuracy and robustness.

CCS Concepts

• **Information systems** → Information systems applications; Data mining.

*Corresponding author



This work is licensed under a Creative Commons Attribution International 4.0 License.

CNSSE 2025, Qingdao, China

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1361-3/2025/02

<https://doi.org/10.1145/3732365.3732409>

Keywords

Anomaly Detection, large language model, Data Semantics, Data Fusion, Cross-domain Reasoning

ACM Reference Format:

Lei Zhang, Jiaxuan Wu, Naiwang Guo, Binbin Wang, Yi Wu, Mengyuan Zhang, and Yinxia Luan. 2025. Anomalous Network Access Detection with Large Language Model. In *2025 5th International Conference on Computer Network Security and Software Engineering (CNSSE 2025), February 21–23, 2025, Qingdao, China*. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3732365.3732409>

1 Introduction

With the rapid development of digital and intelligent technologies, the scale and complexity of power networks are constantly expanding, and the network security risks faced by power systems are also increasing. The key to ensuring system stability and security lies in whether anomalous access behaviors in power networks can be detected in a timely and accurate manner.

Traditional network anomalous behavior detection methods are generally divided into two categories: rule-based detection methods and statistical model-based detection methods^[1]. Rule-based methods mainly rely on artificially constructed rules to identify known anomalous patterns. This method is very effective for detecting known anomalous patterns, but it is difficult to identify unknown or evolving anomalous behavior patterns^[2]. Statistical model-based methods discover potential threats by analyzing data attributes and designing algorithms. Although this method has a certain detection capability for some unknown anomalous patterns, it does not have the ability to process complex high-dimensional data and is not very sensitive to the dynamic changes of anomalous behavior patterns^[3].

With the rise of deep learning and big data technologies, the emergence of machine learning-based detection methods has improved the detection capability of power networks for anomalous

access behaviors to a certain extent^[4]. However, traditional machine learning models still face challenges when processing various unstructured data, especially the difficulty in extracting meaningful relationship features from high-dimensional and heterogeneous data sets. Deep learning models are good at capturing complex features and nonlinear relationships. Although they can make up for the shortcomings of machine learning methods to a certain extent, they still have limitations in detecting complex anomalous patterns in today's network environments^[5]. For example, CNN and RNN have difficulty capturing long-term dependencies^[6], and although Transformer is effective for large-scale log and traffic data, it is difficult to integrate and analyze data from multiple sources^[7].

LLM has become a major breakthrough in the field of natural language processing, demonstrating powerful contextual understanding and reasoning capabilities^[8]. LLM is trained on a large amount of "structured + unstructured" data and is naturally suitable for processing and analyzing multimodal data, such as device logs, network traffic, and sensor data, making it well adapted to The features of modern network systems. In anomaly detection, LLM's unique capabilities in contextual learning and cross-temporal reasoning enable them to capture subtle and progressive attacks^[9], even in the face of previously unseen threats.

Aiming at the shortcomings of existing anomaly detection methods in network systems, this paper proposes an anomaly detection framework based on LLM. By integrating the semantic information of natural language logs, network traffic data, and user behavior data, and leveraging LLM's network security knowledge and cross-domain reasoning capabilities, the model can detect complex and potential anomalies in network systems. The main contributions of this paper are as follows:

- We analyze the strengths and limitations of traditional detection methods and highlight the advantages of LLM in understanding multimodal datasets, reasoning across domains, and detecting complex or potentially anomalous access behaviors.
- We propose a vertical domain LLM-based anomaly detection method. This approach builds on LLM's built-in network security knowledge and cross-domain reasoning capabilities to collaboratively process and analyze information from multiple data sources of network systems for more effective detection of unknown or potentially anomalous access behaviors.
- We validate our approach on a real network system dataset, demonstrating its effectiveness on complex anomalous behavior detection tasks.

The sections of this paper are as follows: section II describes the research problem of this paper, section III details how this paper constructs an LLM-based network anomalous access behavior detection model, section IV presents the experimental results and analysis, and section V summarizes the main work of this paper and proposes future research directions.

2 Preliminaries

The dataset used in this study is derived from system data within a specific power network region, which includes natural language log messages, network traffic data, and user behavior data from

power network systems. We use a pre-trained and fine-tuned cybersecurity vertical domain LLM to understand the multi-source data information and accurately detect anomalous access behaviors in the power network by using the model's built-in cybersecurity knowledge and cross-domain reasoning capabilities in combination with cue learning techniques.

2.1 Problem Definition

Based on the power network system dataset D and the LLM in the field of network security, we comprehensively process and analyze the natural language log information L , the network traffic data F and the user behavior data A in the power network system, and integrate the network security knowledge K and the contextual information U , to achieve the accurate detection of abnormal access behaviors in the power network system. The specific inputs of the model are as follows:

$$f(D, U, K) = (y_t) \forall t \in 1, 2, \dots, n \quad (1)$$

Where $D = \{L, F, A\}$, U represents the context information that affects the detection process, K is the network security knowledge embedded in LLM. y_t is the detection result of the power network access behavior at time t .

3 Methodology

3.1 Dataset Construction

In this paper, we use three datasets to train anomaly detection models for network system access behavior, including a cybersecurity pre-training dataset for the vertical LLM knowledge infusion task, a power network system access dataset for anomalous access behavior detection, and a supervised fine-tuning dataset.

The LLMs for the cybersecurity vertical domain are built by injecting cybersecurity knowledge into the generalized LLMs through continuous pre-training, which leads to the fact that the performance of the LLMs in cybersecurity tasks is closely related to the quality of the cybersecurity pre-training dataset^[10]. Due to the wide range of sources of cybersecurity knowledge, it is important to collect data from sources that contain high-quality or reliable cybersecurity content in order to prevent them from containing incorrect or inaccurate information. Therefore, the sources of pre-trained datasets used in this paper include high-quality security books, security knowledge bases, security research papers, articles from the security community, and vulnerability databases. To avoid noise or redundant data present in the pre-training dataset from negatively affecting model training, this paper also used techniques such as regular expressions to remove corrupted or duplicate data to ensure data integrity and uniqueness, and relevance filtering to ensure that only cybersecurity-related text is retained in the dataset. Ultimately, this paper constructs a high-quality cybersecurity pre-training dataset consisting of approximately 430,000 cybersecurity content items and a total of about 500 million tokens.

For the task of detecting anomalous access behaviors in network systems, the pre-trained LLM needs to be further supervised and fine-tuned to help it accurately understand and recognize complex anomalous access behavior patterns. In this paper, we adopt the method of "data semanticization + expert annotation" to construct a supervised fine-tuning dataset for anomalous access detection. First,

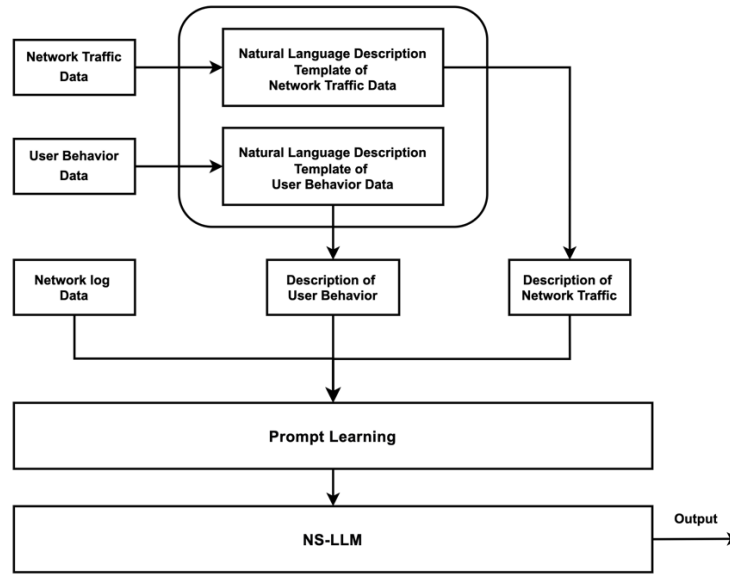


Figure 1: Model Structure Diagram

the structured network traffic dataset and user behavior dataset are semantically processed by describing the original structured data features in natural language, comparing the numerical data with the natural language data; then, the semantic network traffic information, user behavior information and network system log information are learned and reasoned through the powerful closed-source LLM ENIER4.0 to generate and organize anomaly detection results in the form of Q&A. To ensure the scientific rigor of this question-answering dataset, the results are manually annotated by network security experts. The cosine similarity between the manually annotated answers and the answers generated by LLM is calculated, and only the Q&A pairs with similarity scores reaching the threshold are retained. In addition, to avoid overfitting in the supervised fine-tuning process, this paper adopts a strategy of combining anomaly detection data with common sense question-answering data when constructing the supervised fine-tuning dataset. This strategy not only helps to improve LLM’s focus on vertical domain tasks, but also maintains the generalization ability of the model to a certain extent.

The dataset for anomalous access behavior detection in this paper is real power network access data from a specific area, which contains natural language logs, network traffic information, and behavioral data generated by users when interacting with the power network system. By aligning multimodal data with natural language and fusing their semantic features, a power network access dataset is constructed to help the vertical domain language model (LLM) better understand the user’s access behavior pattern, so that the LLM can further use its cross-domain analysis function to more accurately identify anomalous access behavior.

3.2 Model Architecture

The network anomalous access detection framework designed in this paper is shown in Figure 1. The input of the detection framework is multimodal data consisting of natural language log information, network traffic data, and user behavior data. The framework mainly consists of a data semanticization module, a data fusion module, and an anomalous access detection module.

First, the non-natural language data in the multimodal dataset is aligned with the natural language using the predefined natural language description template in the data semanticization module. This converts the multimodal dataset into a multi-source natural language dataset designed for the anomalous access detection task. Next, the multi-source natural language data is integrated into the input of the anomalous access detector based on the data fusion module of prompt learning. Then, the vertical domain LLM for network security is used to capture the anomalous access behavior pattern. This approach enables the framework to effectively extract the semantic information embedded in the multimodal data and integrate the multi-source semantic information through the data fusion module. It makes full use of the security knowledge and cross-domain reasoning ability of the vertical domain LLM to detect anomalous access behavior.

3.2.1 Data Semantics Module. LLMs are pre-trained and fine-tuned based on processed natural language, and their core advantage lies in understanding sentence structure, word relations, and other unstructured text. In their reasoning process, they also rely on natural language context. However, raw structured data (e.g., tables, logs, numerical data) lack explicit natural language context, which means that LLMs may not be able to effectively capture the hidden meanings and relationships between features in structured data. Therefore, natural language descriptions must be used to

Network Traffic Description Template: • Time period description: "At [time period], the system's traffic fluctuated noticeably." • Request type: "Starting from [time point], the requests were mainly [request type], with some [protocol type] requests." • Traffic size: "At [time point], the traffic increased to [traffic size] (unit: traffic unit), significantly higher than the usual traffic size by [comparison percentage]." • Maximum size: "At [time point], the peak traffic reached [maximum traffic size] per second. The largest packet was [maximum packet size] bytes, and the smallest was [minimum packet size] bytes." • Exception handling: "Starting from [time point], an abnormal traffic pattern appeared, and the traffic size reached [traffic size] (comparison percentage with normal traffic), potentially indicating abnormal behavior or attacks." Comprehensive example: "At 2:30 PM, the system traffic increased sharply. The average minute-by-minute traffic increased to 500MB, exceeding the normal rate by 50%. From 192.168.1.2, a large number of GET requests started at 2:35 PM, totaling 1,500 requests, with packet sizes ranging from 512 bytes to 192 KB. This abnormal traffic may indicate a potential DDoS attack."
--

Figure 2: Natural Language Description Template for Network Traffic Data

semantically process complex data features in network systems, thereby providing standardized and universal representations for raw structured data. This promotes the use of LLM's powerful contextual understanding and cross-domain reasoning capabilities.

In the existing power grid system, network traffic data and user behavior data usually include information such as requests, responses, access frequency, and traffic peaks. These data features are highly interpretable and contain rich contextual information, which can be easily converted into natural language by designing text description templates. Therefore, this paper designs corresponding text description templates^[11] for different data sources to effectively explain the meaning of the features of each data set, make the data features more meaningful in the context, and highlight the correlation and overall trend between different features.

The features of the network traffic dataset include time, IP address, request type, number of requests, time point, traffic change, average packet size, maximum packet size, minimum packet size, and anomalous access behavior type. This paper designs corresponding natural language description templates for network traffic data from the perspectives of time period description, request type description, traffic change description, packet size description, and anomalous access behavior description, as shown in Figure 2. In addition, it can be seen from the alignment result example in Figure 2 that the template designed in this paper can accurately reflect the network traffic change pattern in the natural language description. In the subsequent LLM training and reasoning process, these natural language descriptions can better help LLM understand and capture the network traffic change pattern caused by different types of anomalous access behaviors.

Similarly, based on the data features such as user name, login time, IP address, browsed pages, and actions in the user behavior dataset, a natural language description template of the user behavior dataset is constructed from the perspectives of login behavior, page browsing behavior, click behavior, data input behavior, and anomalous behavior, as shown in Figure 3. The example in Figure 3 demonstrates how to use the template to describe the user behavior data features in natural language. The template clearly describes the logic of user behavior and the possible types of anomalous behavior.

LLM can learn the natural language descriptions of different user behaviors and reveal the behavioral patterns that imply different types of anomalous access, thereby improving its ability to detect anomalous access behavior.

3.2.2 Data Fusion Module. The natural language description of network traffic data and user behavior data is combined with the natural language form of network system log information to form a multi-source natural language dataset for anomalous access detection tasks. The fused dataset provides LLM with background knowledge of different dimensions, which helps the model to perform more effective context understanding and cross-domain reasoning^[12], thereby accurately detecting anomalous access behavior.

Prompt learning is an effective strategy for integrating multi-source information^[13]. Its advantage is to design natural language prompts that explicitly guide the model to make use of cross-domain context information for reasoning. In this paper, we design a prompt template for a multi-source natural language dataset for the task of anomalous access detection. By combining network traffic information, user behavior information, and network system log information, the prompt template effectively guides LLM to perform cross-domain reasoning, as shown in Figure 4. From the prompt learning example in Figure 4, the prompt template organizes the natural language descriptions of network traffic, user behavior, and system log information into multiple parts. This structured presentation helps LLM learn and understand different context information. In addition, the prompt also includes a request for LLM to discover the relationship between data from different sources, so that it can comprehensively evaluate whether there is anomalous access behavior.

3.2.3 Anomaly Access Detection Module(NS-LLM). The anomalous access detection module is built on the vertical domain LLM to ensure network security. The input is a multi-source natural language dataset integrated by the data fusion module based on prompt learning. The vertical domain LLM uses the understanding of security domain-specific knowledge to perform cross-dimensional analysis, feature extraction, and reasoning on anomalous access patterns to achieve accurate detection of complex or potentially anomalous access behaviors.

Vertical domain LLMs are optimized for specific domains. During pre-training and fine-tuning, they integrate a large amount of data and tasks from specific domains, and they are able to accumulate deeper domain knowledge compared to general models^[14]. Unlike general LLMs, vertical domain models do not aim to cover a wide range of language tasks, but focus on vertical domains, such as security and anomaly detection in network systems. Vertical domain LLMs for network security integrate a large amount of text resources related to network security, such as research papers, technical documents, vulnerability reports, attack cases, and incident response logs, and accumulate deeper domain knowledge compared to general models. In the task of network security anomaly detection, data usually comes from multiple sources, such as network traffic, user behavior logs, and system alerts. Compared with general LLMs, vertical domain LLMs for network security can align these multimodal data sources and use their powerful

User Behavior Description Template: • Login Behavior: "User [user name] successfully logged into the system at [time], with IP address [IP address], using device type [device type]." • Page Access: "User [user name] visited [page name] at [time], viewing the page for [duration]. The average page viewing time is [average duration]." • Click Behavior: "User [user name] clicked on [element name] on the [page name] page at [time], with [number] total clicks on that page." • Input Behavior: "User [user name] entered [input content] in [input field] at [time], and submitted it." • Download Behavior: "User [user name] downloaded [file name] at [time]. If suspicious download behavior was detected, it may indicate [abnormal behavior]." Comprehensive example: "User John successfully logged into the system at 2:15 PM, with IP address 203.0.113.0, using a Windows device. At 2:25 PM, John visited the homepage, spending 3 minutes on the page, exceeding the average viewing time of 1.5 minutes. At 2:27 PM, John clicked on the 'Submit' button on the contact form. At 2:30 PM, John downloaded a document named 'report.pdf'. After 30 minutes, John logged out of the system, displaying normal login behavior throughout."
--

Figure 3: Natural Language Description Template for User Behavior Data

1. Network Traffic Information "At 2:30 PM, the system traffic increased sharply. The average minute-by-minute traffic reached 500MB, exceeding the normal rate by 50%. From IP 192.168.1.2, a large number of GET requests started at 2:35 PM, totaling 1,500 requests, with packet sizes ranging from 512 bytes to 192 KB. This abnormal traffic may indicate a potential DDoS attack." 2. User Behavior Information "User John successfully logged into the system at 2:15 PM from IP 203.0.113.0 using a Windows device. He visited the homepage for 3 minutes, exceeding the average viewing time of 1.5 minutes. At 2:27 PM, he clicked on the 'Submit' button on the contact form and later downloaded 'report.pdf' at 2:30 PM." 3. System Log Integration "System logs recorded an authentication failure attempt at 2:33 PM, originating from an unknown IP. A security alert was triggered at 2:35 PM due to multiple failed login attempts. Correlating these events, the LLM identifies a potential unauthorized access attempt based on the anomalous traffic spike, abnormal user behavior, and repeated failed authentication attempts." 4. Cross-Modal Correlation Analysis Instruction: • Analyze the relationship between system log entries, network traffic patterns, and user behavior. Identify whether these combined indicators suggest an anomalous access event. • If logs report authentication failures, correlate them with IPs from high-traffic sources to determine potential brute-force attacks. • If a security alert is logged, cross-check the corresponding user session details and network behavior to verify potential intrusion.
--

Figure 4: Multi-Source Natural Language Prompt Template for Anomalous Access Detection

natural language processing and reasoning capabilities to capture the behavioral features of anomalous access and accurately detect network anomalies.

Based on the high-quality network security pre-training dataset and anomalous access detection supervised fine-tuning dataset constructed earlier, this paper further uses the XTuner^[15] training framework to pre-train and fine-tune the basic LLM, injecting vertical domain knowledge in the network security field and enhancing its anomalous behavior detection ability. Considering the efficiency and memory consumption issues during LLM training, this paper uses DeepSpeed^[16] mixed precision training to accelerate calculations, and adopts Flash_Attention^[17] to optimize memory access and calculation order to improve the efficiency of self-attention calculations. In order to obtain the best combination of hyperparameters for pre-training and fine-tuning vertical domain LLM, during the training process, the learning rate, weight decay, optimizer and other hyperparameters are continuously adjusted according to the convergence of the cross entropy loss between the detection results and labels generated by the model and the F1 score indicator on the validation set. The final best hyperparameter combination for pre-training and fine-tuning vertical domain LLM is shown in Tables 1 and 2. In addition, considering that the fine-tuned LLM needs to be used for reasoning in the anomalous access detection task

scenario, the PTH model automatically saved after pre-training and fine-tuning must be further converted to the HuggingFace model format for deployment and reasoning using frameworks such as Xinference.

As the vertical domain LLM learns more patterns and annotations from anomalous access data, it can understand the relationship between data from different sources at a higher level and adaptively adjust the detection strategy and reasoning logic to improve detection accuracy. Therefore, this paper selects the fine-tuned vertical domain LLM for network security as the anomalous access detector and inputs the network traffic, user behavior, and system log information integrated by the data fusion module. The model not only captures the anomalous access behavior existing in a single information source, but also performs cross-domain analysis to detect potential anomalous associations between different information sources, thereby further identifying complex, multi-step anomalous access behaviors. The anomalous access detector designed in this paper will eventually output the detection results of anomalous access. If an anomalous access is detected, the type of anomalous access will be further output to help network security administrators formulate countermeasures.

Table 1: Optimal Hyperparameter Combinations for Pre-training.

Hyperparameter	Learning Rate	Batch Size	Epochs	Sequence Length	Weight Decay	Optimizer
Value	1e-05	256	50	4096	0.01	AdamW

Table 2: Optimal Hyperparameter Combinations for Supervised Fine-tuning.

Hyperparameter	Learning Rate	Batch Size	Epochs	Sequence Length	Weight Decay	Optimizer
Value	2e-05	64	5	512	0.001	AdamW

Table 3: Summary of Baseline Comparison Experiment Results.

model	ACC	Recall	F1-score
InternLM2.5-7B	70.2	65.5	67.8
ChatGLM3-6B	68.5	63.1	65.7
Qwen2-chat-7B	69.8	64.2	66.9
Baichuan2-7B-chat	73.9	69.3	71.5
NS-LLM	85.8	83.3	84.5

4 Experiment

4.1 Datasets and Baselines

This experiment uses real access data from a specific regional power network for training and testing. The dataset contains network traffic, user behavior, and system log information, covering various network activities within 180 days. The data includes both normal access behavior samples and various types of anomalous access. In order to verify the effectiveness of the proposed vertical domain LLM in anomaly detection, we conducted extensive comparative experiments to evaluate its performance in anomaly detection tasks.

We selected several open source general LLMs with the same parameter scale as the baseline model, including InternLM2.5-7B^[18], ChatGLM3-6B^[19], Qwen2-chat-7B^[20], and Baichuan2-7B-chat^[21].

4.2 Anomalous Detection Performance Experiment

In the experiments, we compared the anomaly detection performance of the proposed vertical domain LLM with the baseline model in the task of detecting anomalous access behavior. The input of all models includes natural language descriptions that integrate network traffic, user behavior, and system log information. The evaluation metrics include Accuracy, Recall, and F1-score.

Table 3 summarizes the performance of the proposed method and the baseline model in various evaluation indicators in the same time window. As can be seen from the table, the proposed method performs best in all evaluation indicators, indicating that it is significantly better than other baseline models when dealing with cross-domain reasoning involving multi-source information. At the same time, the excellent performance of the proposed method in F1 score shows that the method can achieve a good balance between precision and recall when capturing anomalous access behavior.

In summary, compared with the existing general LLM of the same parameter scale, the vertical domain LLM shows a significant improvement in cross-domain reasoning ability after pre-training and fine-tuning. In addition, it shows better robustness and adaptability in identifying complex and potentially anomalous access behavior patterns.

5 Conclusion

Aiming at the problem of insufficient accuracy in detecting complex network anomalous behaviors, this study innovatively constructs a multi-dimensional data collaborative analysis framework based on a big language model by fusing network traffic timing features, user operation pattern portraits, and system log semantic information, thus breaking through the limitations of traditional single data source detection methods. Specifically, we inject cybersecurity domain knowledge into the big language model in stages, and sequentially perform pre-training and multi-task supervised fine-tuning on a high-quality cybersecurity corpus to optimize the big language model's ability to recognize anomalous patterns.

The experimental results show that the proposed method has significant improvements in several evaluation metrics compared to a generalized bigram model with comparable parameter sizes, which indicates that the proposed method significantly improves the cross-domain reasoning ability and anomaly detection performance of the bigram model. Meanwhile, the proposed method in this paper also provides new insights and references for future applications of LLMs in the field of cybersecurity, and its future research directions include further optimizing the inference efficiency of vertical-domain LLMs and verifying the method's adaptability to a wider variety of cybersecurity tasks and datasets.

Acknowledgments

This work was supported by Scientific and Technological Project of the State Grid Shanghai Municipal Electric Power Company (grant no. B3094024000J).

References

- [1] Ahmed M, Mahmood A N, Hu J. A survey of network anomaly detection techniques[J]. *Journal of Network and Computer Applications*, 2016, 60: 19-31.
- [2] Patcha A, Park J M. An overview of anomaly detection techniques: Existing solutions and latest technological trends[J]. *Computer networks*, 2007, 51(12): 3448-3470.
- [3] Fernandes G, Rodrigues J J P C, Carvalho L F, *et al*. A comprehensive survey on network anomaly detection[J]. *Telecommunication Systems*, 2019, 70: 447-489.
- [4] Sommer R, Paxson V. Outside the closed world: On using machine learning for network intrusion detection[C]//2010 IEEE symposium on security and privacy. IEEE, 2010: 305-316.
- [5] Cheng, D., Ye, Y., Xiang, S., Ma, Z., Zhang, Y., & Jiang, C. (2023). Anti-money laundering by group-aware deep graph learning. *IEEE Transactions on Knowledge and Data Engineering*, 35(12), 12444-12457.
- [6] Kenton J D M W C, Toutanova L K. Bert: Pre-training of deep bidirectional transformers for language understanding[C]//Proceedings of naacL-HLT. 2019, 1: 2.
- [7] Zhu P, Cheng D, Luo S, *et al*. Leveraging enterprise knowledge graph to infer web events' influences via self-supervised learning[J]. *Journal of Web Semantics*, 2022, 74: 100722.
- [8] Raffel C, Shazeer N, Roberts A, *et al*. Exploring the limits of transfer learning with a unified text-to-text transformer[J]. *Journal of machine learning research*, 2020, 21(140): 1-67.
- [9] Wen Q, Zhou T, Zhang C, *et al*. Transformers in time series: A survey[J]. *arXiv preprint arXiv:2202.07125*, 2022.
- [10] Hassanin M, Moustafa N. A Comprehensive Overview of Large Language Models (LLMs) for Cyber Defences: Opportunities and Directions[J]. *arXiv preprint arXiv:2405.14487*, 2024.
- [11] Li, J., Bian, Y., Wang, G., Lei, Y., Cheng, D., Ding, Z., & Jiang, C. (2023). CFGPT: Chinese financial assistant with large language model. *arXiv preprint arXiv:2309.10654*.
- [12] Miyazawa K, Kyuragi Y, Nagai T. Simple and effective multimodal learning based on pre-trained transformer models[J]. *IEEE Access*, 2022, 10: 29821-29833.
- [13] Liu P, Yuan W, Fu J, *et al*. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing[J]. *ACM Computing Surveys*, 2023, 55(9): 1-35.
- [14] Gu Y, Tinn R, Cheng H, *et al*. Domain-specific language model pretraining for biomedical natural language processing[J]. *ACM Transactions on Computing for Healthcare (HEALTH)*, 2021, 3(1): 1-23.
- [15] XTuner Contributors. Xtuner: A toolkit for efficiently fine-tuning llm. <https://github.com/InternLM/xtuner>, 2023.
- [16] Rasley J, Rajbhandari S, Ruwase O, *et al*. Deepspeed: System optimizations enable training deep learning models with over 100 billion parameters[C]//Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. 2020: 3505-3506.
- [17] Dao T, Fu D, Ermon S, *et al*. Flashattention: Fast and memory-efficient exact attention with io-awareness[J]. *Advances in Neural Information Processing Systems*, 2022, 35: 16344-16359.
- [18] Cai Z, Cao M, Chen H, *et al*. Internlm2 technical report[J]. *arXiv preprint arXiv:2403.17297*, 2024.
- [19] GLM T, Zeng A, Xu B, *et al*. ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools[J]. *arXiv preprint arXiv:2406.12793*, 2024.
- [20] Yang A, Yang B, Hui B, *et al*. Qwen2 technical report[J]. *arXiv preprint arXiv:2407.10671*, 2024.
- [21] Yang A, Xiao B, Wang B, *et al*. Baichuan 2: Open large-scale language models[J]. *arXiv preprint arXiv:2309.10305*, 2023.